



TECHNICAL TASK: AI MODEL TRAINING GUIDE

Project: Smart City Issue Detection System **Role:** AI Engineer **Model:** YOLOv8 Medium (yolov8m.pt)

1. MAIN GOAL

Your task is to train an AI model that can detect city problems from photos. Currently, we will start with **3 Main Problems**:

1. **Pothole** (Tooti hui sadak)
2. **Garbage** (Kachra)
3. **Broken_Streetlight** (Kharab street lights)

Important Note for Future: We might add more problems later (like *Traffic Accidents* or *Fire*). So, keep your folders and dataset very organized. Do not mix everything together.

2. SETUP

You need these accounts (Free):

1. **Roboflow** (For labeling photos).
 2. **Google Colab** (For training the model).
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3. STEP-BY-STEP INSTRUCTIONS

Step 1: Collect Data (Images)

- **Quantity:** Find **200 to 300 photos** for *each* problem.
- **Variety:**
 - Take photos in **Daylight** and **Night**.
 - Take photos from **Near** and **Far**.

- **Organization (Crucial for Future Scaling):**
 - Save images in separate folders on your PC:
 - Folder 1: `raw_data/pothole`
 - Folder 2: `raw_data/garbage`
 - Folder 3: `raw_data/streetlight`
 - *Why?* If we want to add "Fire" later, we will just create a new folder `raw_data/fire` without disturbing the old data.

Step 2: Label Images (Roboflow)

1. Go to **Roboflow.com** and create a project (Select: **Object Detection**).
2. Upload your folders.
3. **Draw Boxes:** Draw a tight box around the problem in every image.
 - Name the classes clearly: `Pothole`, `Garbage`, `Broken_Streetlight`.
4. **Augmentation (Making Model Strong):**
 - Go to "Generate" section.
 - Add **Flip (Horizontal)**.
 - Add **Rotation** (± 15 degrees).
 - Add **Brightness** ($\pm 25\%$).
 - *Reason:* This helps the model work even if the camera is shaking or lighting is bad.
5. **Export:**
 - Click "Generate Version".
 - Select Format: **YOLOv8**.
 - Copy the **Download Code**.

Step 3: Train the Model (Google Colab)

We will use the **Medium Model** (`yolov8m.pt`) because it is smarter than the basic model.

1. Open **Google Colab** (`colab.research.google.com`).
2. Go to Menu: `Runtime > Change runtime type > Select T4 GPU`.
3. Copy and Paste this code into Colab and click **Play**:

Python

1. Install YOLO Library

```
!pip install ultralytics
```

```
from ultralytics import YOLO
```

2. Download Dataset (PASTE YOUR ROBOFLOW CODE BELOW THIS LINE)

Example:

```
# !pip install roboflow
```

```
# ... (Paste code here) ...
```

3. Train the Model (Advanced Settings)
We use 'yolov8m.pt' for high accuracy.
We train for 100 epochs to ensure it learns well.

```
model = YOLO('yolov8m.pt')

model.train(
    data=f'{dataset.location}/data.yaml',
    epochs=100,          # Train for a long time
    imgsz=640,          # Standard image size
    batch=16,           # Good for GPU
    patience=25,        # Stop if no improvement (saves time)
    name='smart_city_final'
)
```

Step 4: Check Results

After training finishes (1-2 hours), check the output folder on the left side: ([runs](#) > [detect](#) > [smart_city_final](#)).

1. Check [confusion_matrix.png](#):
 - This image tells you if the model is confused (e.g., thinking a Pothole is Garbage).
2. Check [results.png](#):
 - Look at the graph. It should be going UP.

4. FINAL DELIVERABLE (Send this to Team Lead)

Go to [runs/detect/smart_city_final/weights/](#) and download:

1. 📁 [best.pt](#) (This is the most important file).
2. 📁 [confusion_matrix.png](#) (Proof of accuracy).
3. 📁 [val_batch0_labels.jpg](#) (Sample usage).

Good Luck!